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A. **Purpose:** To provide guideline instruction for laboratory workflow process.

B. **Scope** : Provision of comprehensive services in the following areas:

a. Hematology
b. Clinical Pathology
c. Biochemistry
d. Serology
e. Immunology

C. **Responsibility**: Pathologist, Laboratory Technician, Doctors (Consultants /Medical Officers), Nurses.

D. **Abbreviations**:

1. OP = Outpatient
2. IP = Inpatient
3. QA = Quality Assurance
4. PT = Prothrombin Time
5. FUS = Fasting Urine Sugar
6. PPUS = Post Prandial Urine Sugar
7. EDTA = Ethylene Diamine Tetra Acetic Acid
8. TRF = Test Requisition Form
9. ABG = Arterial Blood Gas Analysis
10. HCV = Hepatitis C Virus
11. RPR = Rapid Plasma Reagin

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12. CRP = C Reactive Protein
13. ASLO = Antistreptolysin Otiter
14. RA = Rheumatoid Arthritis
15. HbsAg = Hepatitis B virus surface Antigen
16. ER = Emergency Room
17 ICU = Intensive Care Units
18. ICCU = Intensive Coronary Care Unit
19. OT = Operation Theatre

Outsourcing of Laboratory Test:

Laboratory is located in the premises of the hospital but its management is outsourced satisfying the following guidelines:

1. The Laboratory has to comply with the quality standards laid down by the organization.
2. The Laboratory has to ensure confidentiality of patients records and reports
3. The laboratory should have a low turnaround time for reporting of samples.
4. Adequately qualified and trained personnel perform and supervise the investigations.
5. Qualified professionals performing assessment will be registered as applicable under the law of the land:

Professional	Basic Qualification	Registration
Pathologist	M.B.B.S, M.D Path / Microbiology /Biochemistry, DNB Path / Microbiology /Biochemistry	Registered with the Centre / Medical Council.
Laboratory Technicians	DMLT	Not required
Laboratory Attendants	1 Year experience in lab	Not required

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L. <u>List of Test which are outsourced to labs located outside the premises of the hospital</u>:

1. All Test to be performed under Histopathology
2. All Test related to Hormones.
3. All Microbiological Test

Outsourced Lab located outside the premises of the hospital should satisfy the following guidelines:

- The Laboratory has to comply with the quality standards laid down by the organization.
- The Laboratory has to ensure confidentiality of patients records and reports
- The laboratory should have a low turnaround time for reporting of samples.
- Adequately qualified and trained personnel perform and supervise the investigations.
- Qualified professionals performing assessment will be registered as applicable under the law of the land:

Professional	Basic Qualification	Registration
Pathologist	M.B.B.S, M.D Path	Registered with the Centre / Medical Council.
Laboratory Technicians	DMLT	Not required
Laboratory Attendants	1 Year experience in lab	Not required

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E. Departmental Hierarchy:

Department of Laboratory Medicine

Medical Superintendent

Senior Consultant

Laboratory Technicians

Laboratory Attendants

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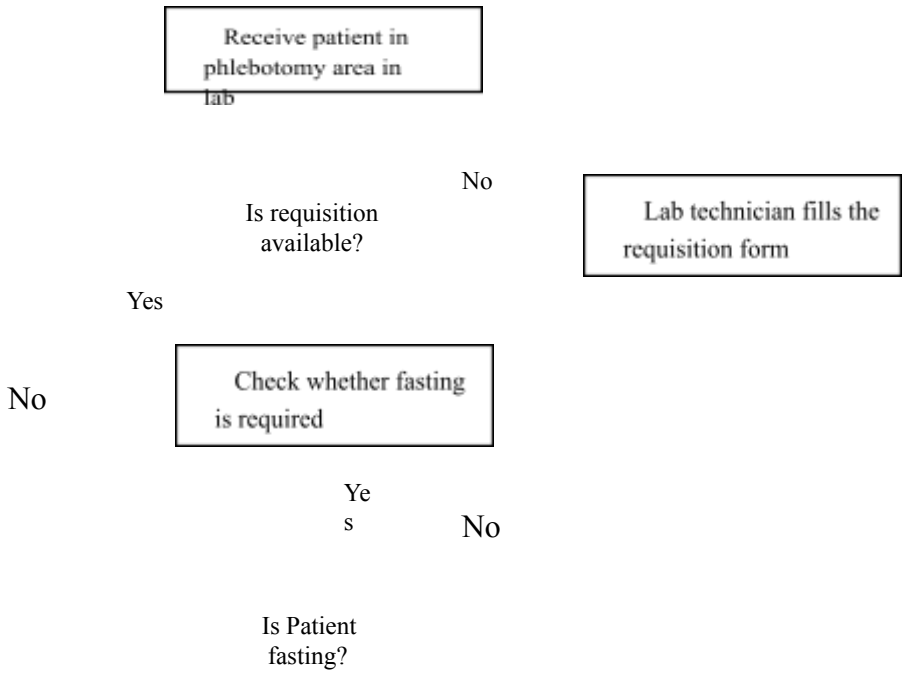
Procedures:

1. Phlebotomy - outpatients



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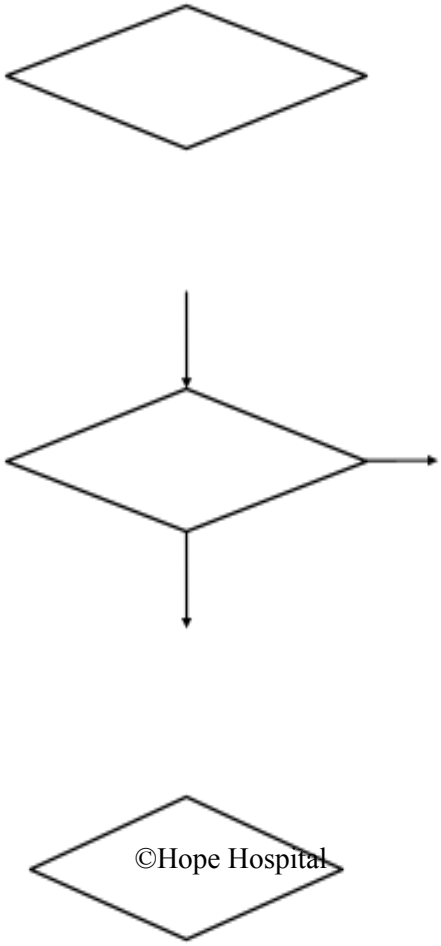

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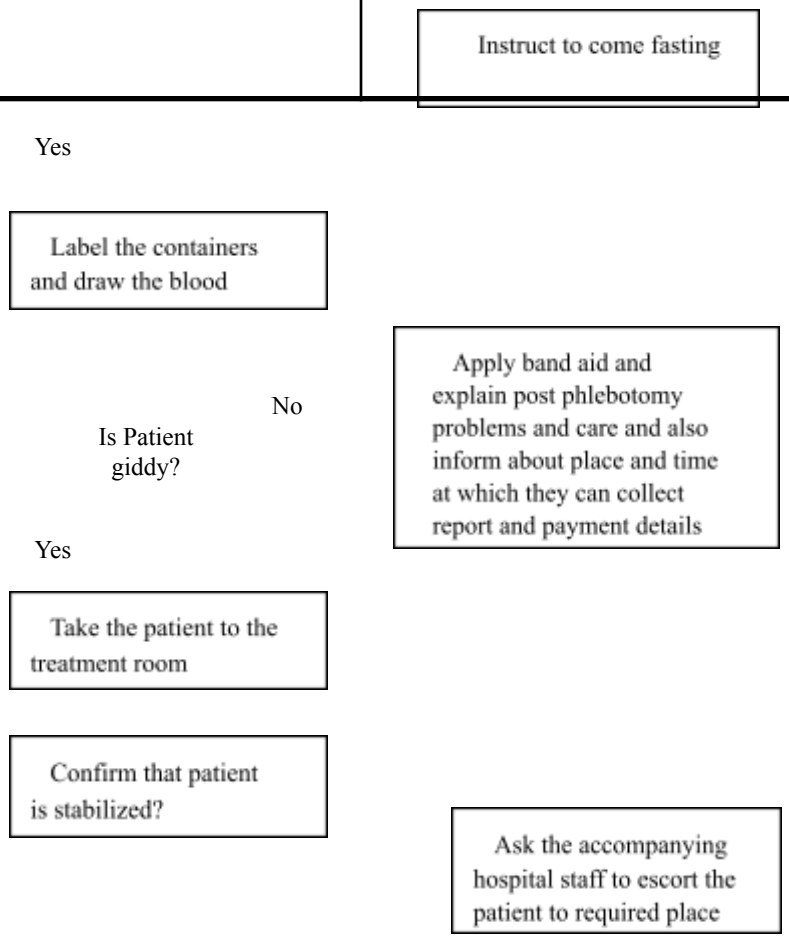

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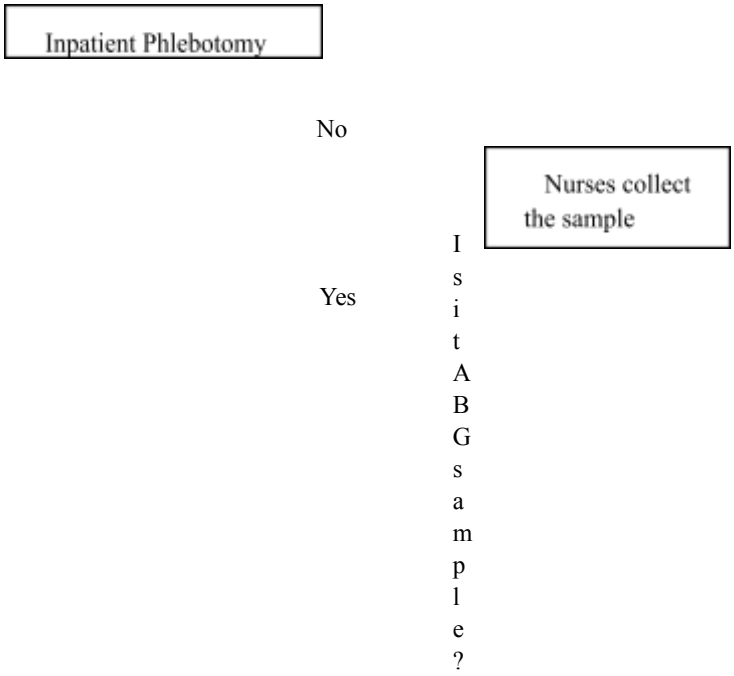
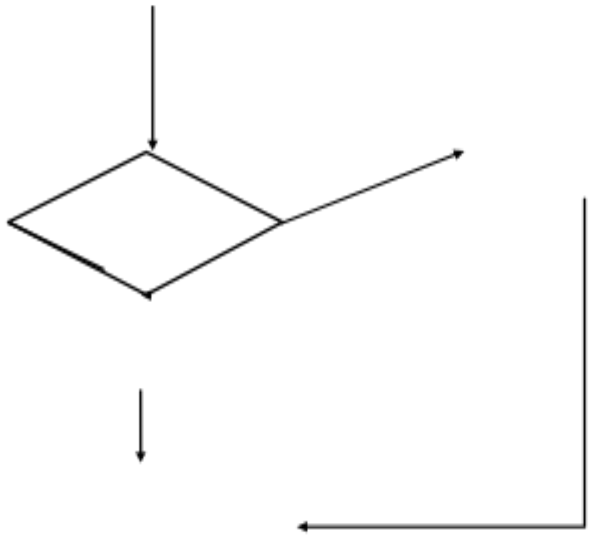
2. Phlebotomy – Inpatients

In patients phlebotomy is carried out in the wards and is usually done by the nurse





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MO or Anesthetist collects the sample

Labeled Samples with requisition are sent to lab for analysis

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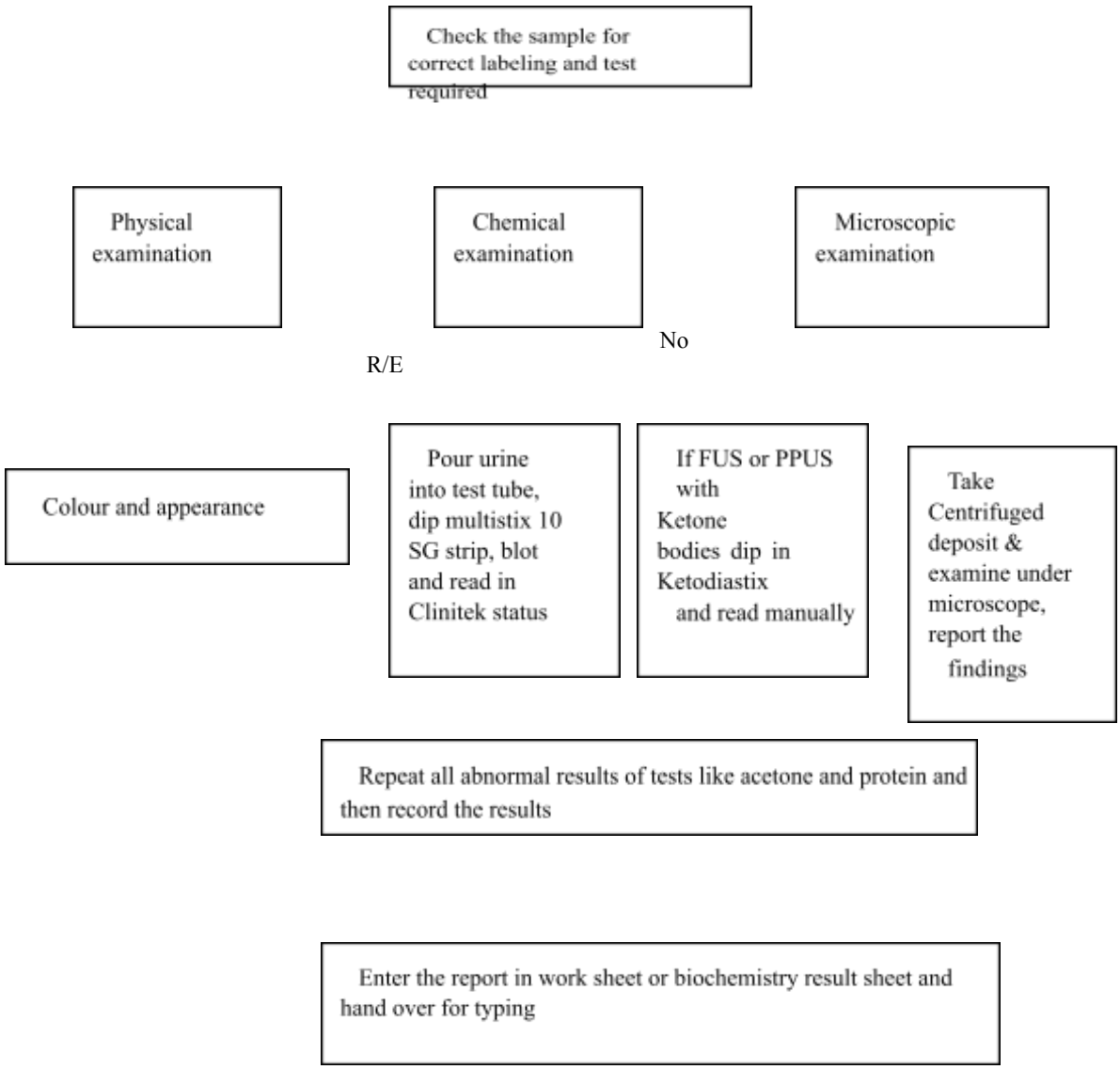
3. **Clinical Pathology**:

Urine **Analysis**



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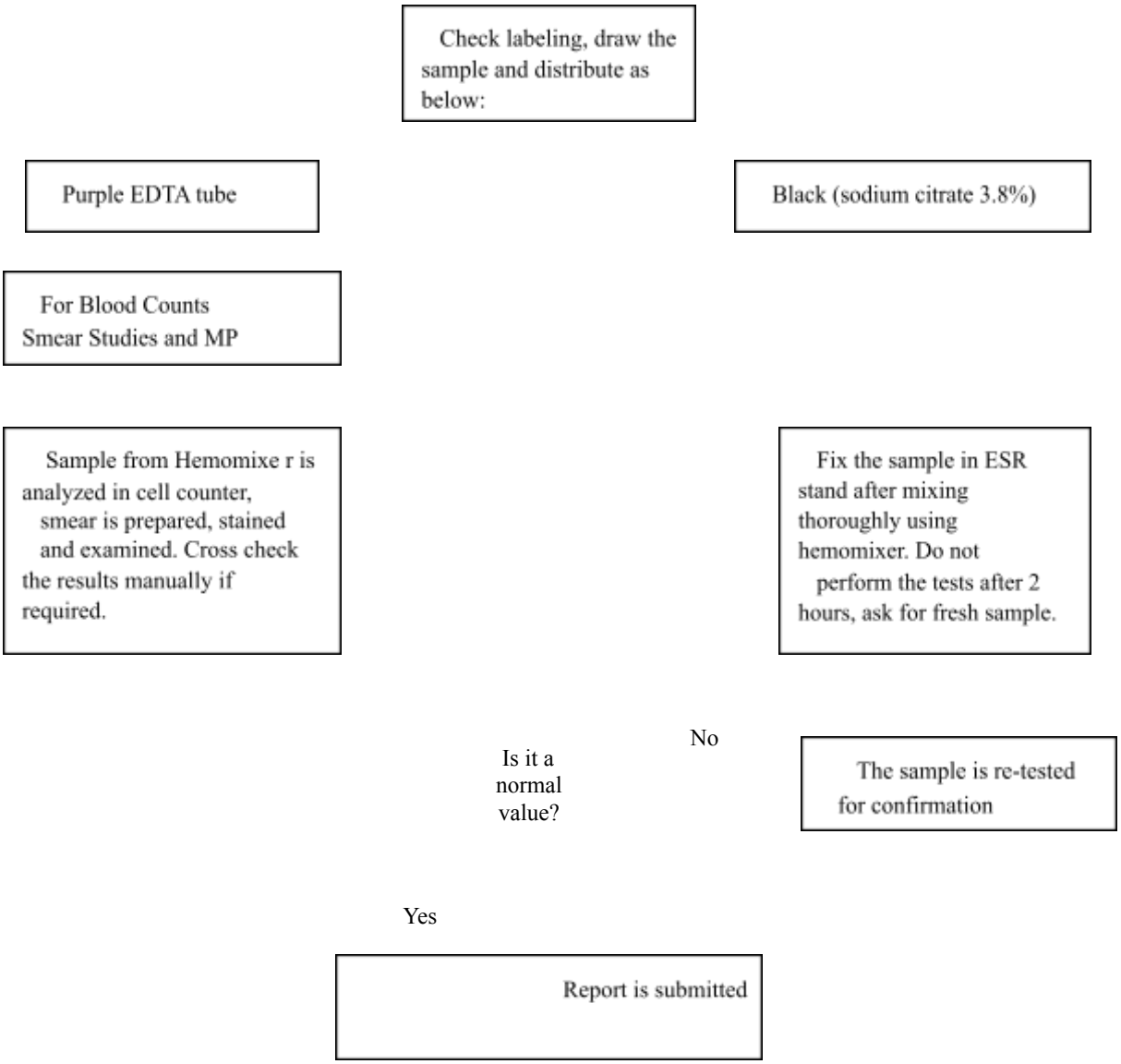
4. Hematology:



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5. Biochemistry:

Cross check the sample label against requisition

Check for hemolysis, gross lipemia, & hyperbilirubinemia

Yes

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No

Programme the instrument after transferring the required amount of sample in sample cups with label written by marker pen

No

Is it Normal?

Repeat the test

Make a note of the same on TRF

Yes

Send sample and TRF to the next section for analysis

Report the result

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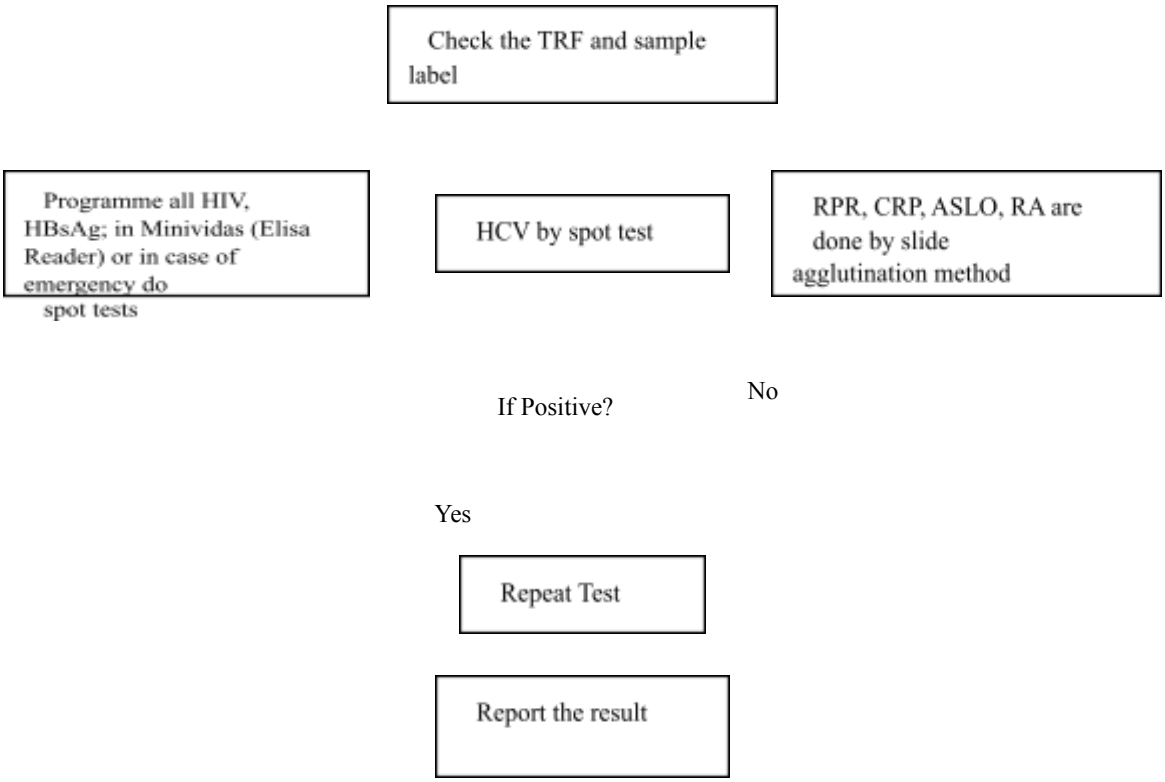
6. Serology:



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7. Disposal of Specimen:

Nature of the sample	
Blood samples collected in vacuum tubes (Vaccutainer)	Urine and stool samples are taken in Sterile container
Discard after seven days of collection into Blue bin	Urine samples are disinfected by hypochloride solution for 10 minutes and then discarded in lab sink; the container is rinsed in tap water and immersed in detergent

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G. Departmental Policies:

a. Outpatients:

Sample Collection and performance of the test:

1. Treating Doctor prescribes the required laboratory test in the OP case sheet of the patient and fills a laboratory test requisition slip.

2. Patient /relatives visit the cash counter with the laboratory test requisition slip for payment of the user charges for the prescribed test.

3. The clerk collects the user charges for the prescribed test as per indicated in the laboratory test requisition slip and enters the details in the designated register and assigns a patients sample identification number to the patient .The number is entered in the laboratory test requisition slip of the individual patient.

4. Receipt for the payment of the user charges are handed over to the patient / relatives by the cashier.

5. Patient visit the sample collection area of the laboratory department and submits the laboratory test requisition slip along with the payment receipt to the laboratory attendant.

6. The patients details including the name of the patient , age , sex , patients sample identification number , tests to be performed , referring doctor are entered in the record register.

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7. Patients are requested to wait in the waiting area of the department till they are called. Patient's sample collection is done strictly on the basis of "FIRST COME FIRST SERVE".

8. Patients are called in the sample collection area of the department as per their turn. The Laboratory technician collects the patient's sample, stores the same in leak proof containers labels the container with the patient's name, patients sample identification number and sample type.

9. Patients are informed about the time and day for collecting the test reports

10. The collected sample is forwarded to the main laboratory area for the performance of the prescribed test.

11. Test are performed and reports generated .All reports are signed ,dated clearly by authorized laboratory personnel.

12. Patients/relatives collect the report from the reception area of the laboratory at the mentioned time and day.

b. Inpatient:

Sample Collection and performance of test:

1. The treating doctor prescribes the requisite laboratory test in the patient's case sheet, fills the laboratory test requisition slip and forwards the same to the ward nurse.

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2. Patients relatives are informed about the test to be performed by the ward nurse. Ward nurse directs the patients relatives to the cash counter of the hospital for the payment of the user charges.

3. Patients relatives visit the cash counter along with the test requisition slip. The clerk collects the user charges for the prescribed test as per indicated in the laboratory test requisition slip and enters the details in the designated register and assigns a patients sample identification number
.The number is entered in the laboratory test requisition slip of the individual patient

4. The Ward nurse collects the patient's sample, stores them in leak proof containers, labels the sample with patient's name, age, sex, sample type, patients sample identification number as indicated in the laboratory test requisition slip.

5. The ward nurse informs the laboratory for collection of the sample .The laboratory attendant collects the patients sample along with the test requisition form from the inpatients wards.

6. Samples are received in the laboratory, records relating to the name of the patient, age ,sex , sample type and patients sample identification number are entered in the record book.

7. Samples are forwarded to the main laboratory area for processing.

8. Test are performed and reports generated .All reports are signed, dated clearly by authorized laboratory personnel.

9. The reports are forwarded directly to the staff nurse of the designated inpatient care area

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from where patient's sample was collected.

H. Identification, Handling and Transportation of samples :

1. A**l** samples are properly labeled with the patient's name, unique patient sample identification number, sample type and patient type ie.OP or IP, incase IP the specific ward from where the patient's sample has been collected is also indicated in the label.
2. Sample without proper label as per the guideline stated above would be rejected .In such case the sample is returned to the specific area from where it has been collected.

3. Vacuum containers are used for transfer of samples from the specific point of collection to the main laboratory area so that there is no spillage .Samples which are to be outsourced are separated into serum or plasma as required, sample bullets prepared, labeled, record entered in the outsource test register and then transferred to the specific outsourced lab. All the vacuum containers are placed in a sample carrier box to avoid spillage and safe transport to lab.
4. Staff responsible for handling of patient samples is required to wear appropriate personal protective equipment which include gloves, aprons, face mask.

c. Reporting of Urgent Cases:

Urgent cases pertains to such patients who has to undergo immediate laboratory investigation incase the treating doctor suspect any form of medical emergency which can be confirmed by the prescribed laboratory investigation.

In such cases the following procedures are followed:

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1. In case laboratory test are to be performed on an emergency basis the same is indicated in the laboratory test requisition slip filled by the doctor.

2. The concerned patient's samples are labeled "URGENT", along with the other routine details.

3. Sample are collected from the designated area (laboratory sample collection area / inpatient care area of the hospital) as applicable and immediately forwarded to the main laboratory area for testing
4. Test performed and results obtained.

5. Results are immediately intimated directly to the treating doctor , incase the primary treating consultant of the patient is not available the result of the test are brought into the notice of the available medical officer. Test results are also notified to the concerned ward nurse of the patient , in case the patient is admitted in the hospital.

6. The name of the person who has been intimated about the test result are documented along with the time and date.

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d. Reporting of critical test results:

Critical results are those result values which require immediate attention by the doctor/nurse failing which there is a danger of harm to the patient.

This policy prevents the critical result from being lost in communication failures, improves patient safety, and documents the delivery of the results. . The goal is to improve patient safety by encouraging hospitals and their staff to voluntarily report events that adversely affected patients. Ineffective communication is the most frequently cited root cause for sentinel events. Hope hospital has a standardized approach to hand-off communications, including an opportunity to ask and respond to questions. This saves the medical organization from communication errors that could result in patient injury or death, and lawsuits against the caregiver.

Policy for reporting such result values are as follows:

When a radiologist, pathologist, interpreting clinician, or emergency department or laboratory personnel flag a study as a critical finding, this critical information is sent immediately to a healthcare professional (such as surgeons, physicians, nurses, etc.) telephonically. This information can include images, reports, annotations, handwritten notes, and other critical information. When the recipient receives the message a receipt confirmation is noted in the register maintained in the lab or imaging unit. Audit trails ensure that hospital administration has a way to document and check that results have gotten to an appropriate caregiver in (1 hour, 8 hours or 3 days (“Red”, “Orange”, “Yellow” time targets differ!) All require acknowledgment in target time frames.

1. Who the results should go to?

   The report should reach a provider who can take action and is documented.

   Make commitment to call the consultant directly for “red” alerts all laboratory test results, which are so far from the reference range that they indicate a potentially dangerous condition requiring immediate attention, will be intimated to the concerned consultant forthwith.

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2. Who the results should go to when the ordering provider is not available?

In case the consultant is not reachable the result will be brought into the notice of the Medical Officer on duty and is documented.

3. The result will also be informed to the concerned ward nurse if the patient is admitted to hospital and is documented.

This policy applies to Inpatients, outpatients, and emergency settings This policy applies to Laboratory, radiology and other diagnostic tests

What are critical tests and what test results require timely and reliable communication?

Defining critical values/interpretations for all diagnostic areas. List includes, but is not limited to…

New diagnosis of…If represents a change of…

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Appendix 2. Excerpt from Consensus Group Recommendations for "Starter Set" Values/Interpretations for the Red, Orange, and Yellow Categories from Laboratory, Cardiology, and Radiology*

Testing Area	Alerting Category	Test	Red Category Conditions Complete Alert within 1 hour	Orange Category Conditions Complete Alert within 8 hours May be acceptable to defer alert between 11pm and 6am	Yellow Category Conditions Complete Alert within 3 days
Chemistry	Always Red (Always potentially life-threatening to patient, even if patient is actively being treated)	Glucose (adult)	HIGH (e.g. > 400) LOW (e.g. < 50)		
		Potassium	HIGH (e.g. > 6) LOW (e.g. < 2.8)		
		Phosphorus	LOW (e.g. < 1.0)		
		Sodium†	HIGH (e.g > 160) LOW (e.g. < 120)		
		Bicarbonate‡	LOW (e.g. < 10)		
	Red on first instance, orange thereafter (Most dangerous when first detected, but providers should know about these results as the condition is being treated)	Bicarbonate‡	LOW (e.g. 10–15), first instance only§)	LOW (e.g. 10–15), not first instance§)	
		Magnesium	HIGH (e.g. > 5), first instance only§. May tolerate higher level for L&D patients LOW (e.g < 1), first instance only§	HIGH (e.g. > 5), not first instance§. May tolerate higher level for L&D patients LOW (e.g < 1), not first instance§	
		Calcium (total or ionized)	HIGH (e.g. >13 total, >6 ionized), first instance only§ LOW (e.g. < 7, < 3.5 ionized total), first instance only§	HIGH (e.g. > 13 total, > 6 ionized), not first instance§ LOW (e.g. < 7 total, < 3.5 ionized), not first instance§	
		CK/MB	HIGH, indicative of acute MI, first instance only§		
		Troponin	HIGH, indicative of acute MI, first instance only§		
	Orange on first instance, routine thereafter (Only worth alerting once so that providers can initiate	Lactic Acid	HIGH (e.g. > 5), first instance only§	HIGH (e.g. > 5), not first instance§	
		SGOT		HIGH (e.g. > 500), first instance only§	
		SGPT		HIGH (e.g. > 500), first instance only§	
		BUN		HIGH (e.g. > 100), first instance only§	

Appendix 2. Excerpt from Consensus Group Recommendations for "Starter Set" Values/Interpretations for the Red, Orange, and Yellow Categories from Laboratory, Cardiology, and Radiology*

Testing Area	Alerting Category	Test	Red Category Conditions Complete Alert within 1 hour	Orange Category Conditions Complete Alert within 8 hours May be acceptable to defer alert between 11pm and 6am	Yellow Category Conditions Complete Alert within 3 days
Chemistry	Always Red (Always potentially life-threatening to patient, even if patient is actively being treated)	Glucose (adult)	HIGH (e.g. > 400) LOW (e.g. < 50)		
		Potassium	HIGH (e.g. > 6) LOW (e.g. < 2.8)		
		Phosphorus	LOW (e.g. < 1.0)		
		Sodium†	HIGH (e.g > 160) LOW (e.g. < 120)		
		Bicarbonate‡	LOW (e.g. < 10)		
	Red on first instance, orange [illegible]		LOW (e.g. 10-15), first instance [illegible]	LOW (e.g. 10-15), not first [illegible]	
	Orange on first instance, routine thereafter (Only worth alerting once so that providers can initiate treatment program within the next few hours)	Lactic Acid	HIGH (e.g. > 5), first instance only§	HIGH (e.g. > 5), not first instance§	
		SGOT		HIGH (e.g. > 500), first instance only§	
		SGPT		HIGH (e.g. > 500), first instance only§	
		BUN		HIGH (e.g. > 100), first instance only§	
		Creatinine		HIGH (e.g. > 4), first instance only§	
		Amylase		HIGH (e.g. > 500), first instance only§	
		Lipase		HIGH (e.g. > 200), first insance only§	
	Yellow only (Rarely an emergency, but needs follow-up within days)	TSH			HIGH (e.g > 50) LOW (e.g. < 0.1)
		Lead			HIGH (e.g. > 25)
Blood Gases‖	Always Red	pH	HIGH (e.g. > 7.6) LOW (e.g. < 7.2)		
		pO_2	LOW (e.g. < 60)		

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H.Identification, Handling and Transportation of samples:

5. A**l** samples are properly labeled with the patient's name, unique patient sample identification number, sample type and patient type ie.OP or IP, incase IP the specific ward from where the patient's sample has been collected is also indicated in the label.
6. Sample without proper label as per the guideline stated above would be rejected .In such case the sample is returned to the specific area from where it has been collected.
7. Vacuum containers are used for transfer of samples from the specific point of collection to the main laboratory area so that there is no spillage .Samples which are to be outsourced are separated into serum or plasma as required, sample bullets prepared, labeled, record entered in the outsource test register and then transferred to the specific outsourced lab.
8. Staff responsible for handling of patient samples is required to wear appropriate personal protective equipment which include gloves, aprons, face mask.

I.Turn Around Time:

For routine investigations

Haemogram :- 60 minutes

Biochemistry like blood urea, Sr. Creatinine, glucose, SGOT, SGPT - 45 minutes HIV tridot , HBsAg by rapid testing:- 45 minutes

Arterial blood gases :- 45 minutes

Urine R – 30 minutes

For emergency investigations

Haemogram – 20 minutes

Biochemistry like blood urea, Sr. Creatinine, glucose, SGOT, SGPT - 20

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minutes HIV tridot , HBsAg by rapid testing:-20 minutes

Arterial blood gases :- 25 minutes

Investigation request received from critical care areas of the hospital like Emergency, ICU, ICCU, OT etc would be given priority.

J. Disposal of Specimen:

1. All specimens are discarded as per the biomedical waste management policy of the Hospital.
2. Urine and stool samples are discarded every day.
3. All blood samples are discarded after seven days.
4. Used or expired blood bags are discarded in to yellow bins.
5. All isolates and used media are disinfected using sodium hypocholride solution and discarded into yellow bin.

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